

Linux Programming Laboratory Record

1. Arithmetic Operation Program

Aim: To perform arithmetic operations using shell scripting.

Procedure:

1. Open the terminal in Linux.
2. Create a shell script using nano editor.
3. Enter commands for addition, subtraction, multiplication and division.
4. Save the script file with .sh extension.
5. Provide execute permission using chmod.
6. Run the script in terminal.
7. Display arithmetic calculation results.
8. Verify the generated output.

Result: Thus, arithmetic operations were performed successfully.

2. Store Bill Processing with GST

Aim: To calculate store bill amount with GST using shell scripting.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Enter product quantity and price details.
4. Write GST percentage calculation commands.
5. Save the script file.
6. Execute the script after permission change.
7. Display subtotal, GST and final amount.
8. Verify the calculated bill output.

Result: Thus, the store bill with GST was generated successfully.

3. Employee Payroll System

Aim: To prepare employee payroll processing using shell script.

Procedure:

1. Start the Linux terminal.
2. Create a payroll shell script.
3. Enter employee salary and allowance details.
4. Write commands to calculate gross salary.
5. Include deduction calculations.
6. Save and execute the program.
7. Display employee payroll details.
8. Verify the salary output.

Result: Thus, the employee payroll system was executed successfully.

4. Student Mark List using Result and Grade

Aim: To prepare student mark list with grade calculation.

Procedure:

1. Open the terminal window.
2. Create a shell script for mark processing.
3. Enter marks for different subjects.
4. Calculate total and average marks.
5. Write conditions for result and grade.
6. Save the shell program.
7. Execute the script file.
8. Verify grade and result displayed.

Result: Thus, the student mark list and grade were generated successfully.

5. Electricity Board Bill Processing

Aim: To calculate electricity bill amount using shell script.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Enter electricity unit consumption.
4. Write tariff calculation commands.
5. Include additional charge calculations.
6. Save the program file.
7. Execute the shell script.
8. Verify the electricity bill amount.

Result: Thus, electricity bill processing was completed successfully.

6. Find the Biggest Number

Aim: To find the biggest number among given values.

Procedure:

1. Open terminal in Linux.
2. Create a shell script file.
3. Read input numbers from user.
4. Use conditional statements for comparison.
5. Find the largest value.
6. Save the script file.
7. Execute the program.
8. Display the biggest number.

Result: Thus, the biggest number was identified successfully.

7. Calculate the Sum of Numbers

Aim: To calculate the sum of numbers using shell scripting.

Procedure:

1. Open the Linux terminal.
2. Create a shell program.
3. Read numbers from the user.
4. Initialize sum variable.
5. Use arithmetic operators for addition.
6. Save the shell script.
7. Execute the program.
8. Display the total sum.

Result: Thus, the sum of numbers was calculated successfully.

8. Compute Factorial Value

Aim: To compute factorial value using shell script.

Procedure:

1. Open terminal window.
2. Create a shell script file.
3. Read the number from user.
4. Use loop statements for multiplication.
5. Calculate factorial value.
6. Save the program file.
7. Execute the shell script.
8. Verify the factorial output.

Result: Thus, factorial value was computed successfully.

9. Print the Fibonacci Series

Aim: To generate Fibonacci series using shell scripting.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Initialize Fibonacci variables.
4. Read the limit value from user.
5. Use loop statements to generate series.
6. Save the program file.
7. Execute the shell script.
8. Display Fibonacci sequence.

Result: Thus, the Fibonacci series was printed successfully.

10. Simple Calculation Using Switch-Case

Aim: To perform calculations using switch-case statement.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Read two numbers and operator.
4. Use switch-case for operation selection.
5. Perform arithmetic calculations.
6. Save the script file.
7. Execute the shell program.
8. Verify the displayed result.

Result: Thus, calculations using switch-case were executed successfully.

11. Calculate the Sum of Numbers (CSH)

Aim: To calculate sum of numbers using C shell scripting.

Procedure:

1. Open C shell terminal.
2. Create a C shell script file.
3. Enter input values.
4. Initialize sum variable.
5. Write addition commands.
6. Save the program.
7. Execute the C shell script.
8. Display the calculated sum.

Result: Thus, the sum was calculated successfully using C shell.

12. Compute Factorial Value (CSH)

Aim: To compute factorial using C shell scripting.

Procedure:

1. Open Linux terminal.
2. Create a C shell script.
3. Read the input number.
4. Use looping statements.
5. Multiply values repeatedly.
6. Save the shell script.
7. Run the C shell program.
8. Display factorial result.

Result: Thus, factorial value was computed successfully using C shell.

13. Print the Fibonacci Series (CSH)

Aim: To print Fibonacci series using C shell.

Procedure:

1. Open the terminal window.
2. Create a C shell script file.
3. Initialize Fibonacci variables.
4. Enter limit value.
5. Use loops for series generation.
6. Save the script file.
7. Execute the shell program.
8. Display Fibonacci output.

Result: Thus, Fibonacci series was generated successfully using C shell.

14. Find the Armstrong Number

Aim: To check whether a number is Armstrong or not.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Read the input number.
4. Separate digits using arithmetic operations.
5. Calculate cube sum of digits.
6. Compare with original number.
7. Execute the shell script.
8. Display Armstrong result.

Result: Thus, the Armstrong number was identified successfully.

15. Prime Number Generation

Aim: To generate prime numbers using shell scripting.

Procedure:

1. Open Linux terminal.
2. Create a shell script file.
3. Enter the range value.
4. Use loops for divisibility checking.
5. Identify prime numbers.
6. Save the program file.
7. Execute the shell script.
8. Display prime numbers.

Result: Thus, prime numbers were generated successfully.

16. Process Commands

Aim: To study Linux process management commands.

Procedure:

1. Open Linux terminal.

2. Enter process commands like ps and top.
3. Execute the commands.
4. Observe running process details.
5. Use kill command for process control.
6. Monitor CPU usage.
7. Verify command outputs.
8. Analyze process information.

Result: Thus, process management commands were studied successfully.

17. Files and Directories Commands

Aim: To study Linux file and directory commands.

Procedure:

1. Open Linux terminal.
2. Create files and directories.
3. Use ls command to list files.
4. Perform copy and move operations.
5. Rename files and folders.
6. Delete unwanted files.
7. Execute directory navigation commands.
8. Verify command outputs.

Result: Thus, file and directory commands were executed successfully.

18. Text Processing Commands

Aim: To study Linux text processing commands.

Procedure:

1. Open Linux terminal.
2. Create a sample text file.
3. Use cat command to display contents.
4. Apply grep for searching text.
5. Use sort and wc commands.
6. Execute cut and tr commands.
7. Observe processed text output.
8. Verify the command results.

Result: Thus, text processing commands were executed successfully.

19. Memory Management Commands

Aim: To study memory management commands in Linux.

Procedure:

1. Open Linux terminal.
2. Execute free command.

3. Check memory usage statistics.
4. Use top command for monitoring.
5. Observe swap memory details.
6. Analyze used and free memory.
7. Execute memory related commands.
8. Verify displayed memory information.

Result: Thus, memory management commands were studied successfully.

20. Date and Time Commands

Aim: To study date and time commands in Linux.

Procedure:

1. Open Linux terminal.
2. Execute date command.
3. Display current system date.
4. Display current system time.
5. Use cal command for calendar view.
6. Modify date format if required.
7. Observe command outputs.
8. Verify date and time information.

Result: Thus, date and time commands were executed successfully.